

Full Stack Software Developer

Versatile full stack software developer with an emphasis on front end development with React, Electron, and Typescript, and backend development with various [Node.js](#) frameworks and Python. Comfortable working with both relational and non-relational databases as well as dev ops. Additional experience working on embedded platforms in C and C++, and .NET applications, Golang, and various front and backend typescript frameworks.

Work Experience

Software Developer

Annaly Capital Management (October 2021 - Present)

- Develop internal web and desktop applications using React, Redux, and Electron to assist our trading and treasury team.
- Switched to lazy loading for application resources which reduced start up times by up to 25%.
- Developed REST APIs using python and Flask to support desktop and web applications
- Created views, tables and stored procedures using MS SQL Server.
- Designed and created applications in React to replace a series of excel sheets, allowing the treasury team to save time managing daily excel reports, and improved data consistency.
- Translated user requirements into technical requirements and deliverables for myself and my team.
- Debugged and assisted in the development of back end services using .NET.
- Worked with Curium master data management system to validate and master data from multiple sources.
- Worked with multiple UI component libraries (Material UI, AG Grid) as well as CSS and Tailwindcss.
- Collaborated remotely with a team based in New York and in person with coworkers in Toronto.
- Developed standards for git usage and typescript to improve consistency between developers.
- Mentored co-op students, reducing the ramp up for learning our systems.

Electronics Software Developer

Leggett & Platt Automotive. (October 2018 - September 2021)

- Developed software for embedded systems in C using the STM8 platform.
- Redesigned internal testing applications from the ground up using React and Electron. Used Electron's IPC to get information from devices via UART, and create logs. Reduced hardware needed for testing by over 75% by supporting multiple connections to our testing devices.
- increased communication speed of our embedded controller by 800%, allowing for more robust logging between messages.
- Redesigned demo application used by major OEMs for demoing our systems
- Redesigned message pattern building app to allow customers to create, save and export message patterns to our system. Reduced load times over previous generation applications by over 90%. Increased message patten upload speeds by 800%.
- Developed tools for analyzing data using python which greatly improved our test log analysis.
- Derived requirements from stakeholders to system and software requirements.
- Support and oversee tests for company products in our internal test lab and at our manufacturing facilities.

- Implemented git flow to standardize the software teams usage of git.
- Assisted in software architecture design and creation of a standard coding ruleset.

Co-op Work Experience

Radix Inc. (September 2017 - January 2018)

- Updated UI of WPF applications using XAML
- Developed factory line simulation using Unity3D and C#
- Implemented A* pathfinding algorithms in simulation
- Unit tested code using NUnit.

Brave Control Solutions (May 2016 - August 2016 and January 2017 - May 2017)

- Created automated parking garage simulation using Unity3D and C#
- Worked with NodeJS servers to simulate PLC communication with simulations
- Worked with PLCs to control simulated machines
- Learned model design using Blender and texture design using Substance Designer

Skills

- TypeScript / JavaScript (React, Vue, Svelte, Redux, Zustand, Electron, Next.js, Express, Elysia, Jest, React Testing Library, Vitest, TailwindCSS, MUI) - 8 years
- Git - 8 years
- Go (Gin, Fiber, Gothic) - 1 year
- Python (Flask) - 5 years
- C (Embedded C for STM8) - 3 years
- C# (WPF, Unity3D, .NET) - 3 years
- C++ (Unreal Engine 4 / 5, embedded C++) - 2 year
- Swift (iOS development, Swift UI) - 0.5 years
- SQL (SQL Server, Postgres) - 4 years
- Docker - 2 years
- No-SQL (MongoDB) - 1 year

Comfortable learning and working with any language and tech stack.

Education

Bachelor of Applied Science, Electrical Engineering with Co-op 2014-2018

University of Windsor, Windsor, Ontario

- Ontario Entry Scholarship, Kerry Inc. Scholarship

References available upon request